

Comprehensive energy poverty index: Measuring energy poverty and identifying micro-level solutions in South and Southeast Asia

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ABSTRACT

The absence of adequate modern sources of energy impedes the presence of decent living conditions. This very lack of modern sources of energy is Energy Poverty. Lack of availability, accessibility, and affordability are the main reasons behind this problem. In the literature, all three aspects were not covered together. In this paper, we have developed a composite index of energy poverty to measure the magnitude of energy poverty in the ASEAN region and India, where we capture all three elements. The measure revealed the degree of problem in each country, with Cambodia being the most energy poor amongst the countries in the study and Thailand being the least energy poor. To dig deeper into the problem at the micro level and to deliberate the policy implications, we developed detailed village-level case studies and analysed region-specific energy poverty. We thus identify the key root causes and propose effective solutions to eradicate the prevailing problem in these regions.

1. Introduction

Energy is essentially required for basic human needs particularly lighting, heating, communication, basic automation and public facilities,¹ besides different commercial activities that generate necessary income. If there is a lack of sufficient, consistent, fine quality, environmentally non-threatening energy sources for these activities; it is termed as energy poverty (Reddy, 2000; Access to Energy in Developing Countries, 2002). The UN's Millennium Development Goals – whose purpose is to eliminate severe poverty, improve living conditions and enable progress towards sustainable development, do not include any mention of energy poverty (González-Eguino, 2015; Gwénaëlle et al., 2009). However, the statistics give some disquieting facts. Approximately 1.1 billion worldwide do not have electricity as per IEA special report (IEA, 2017) and 107 million out of this population live in the ASEAN region. In the ASEAN region, about 50% are in Indonesia, followed by Myanmar and the Philippines. Only four countries (Brunei, Malaysia, Singapore, and Viet Nam) have almost 100% electrification. Moreover, 3.04 billion people are dependent on ‘traditional biomass’²

for their cooking needs according to World Bank's Global Tracking Framework 2017 (World Bank, 2017). Two-thirds of these people live in Asia, and to a lesser extent Africa, where clean cooking is still not a policy priority. 880 million out of 3.04 billion (29%) are in the ASEAN region. Lack of access of clean cooking fuels results in indoor and outdoor air pollution causes about 3.8 million deaths each year worldwide (WHO, 2018).³

The above statistics refer to lack of accessibility to energy. However, there is another reason of energy poverty, i.e. unaffordability. Energy poverty exists primarily because either (i) people do not have access to energy (Rao and Pachauri, 2017), (ii) energy is not available, or (iii) people cannot afford it (IAEA, 2005). Lack of infrastructure, investment (Khandker et al., 2012; Birol, 2007) along with an inability to capture remote or low population density areas act as impediments to accessibility and availability of energy to people. Whereas, energy is non-affordable mainly due to lack of income or wealth (Heindl and Schussler, 2018), high energy prices, or high cost of supplying modern energy sources (Middlemiss and Gillard, 2015; Khandker et al., 2012). On the basis of these categories, there are three groups of people which come

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¹ Such as provision of education, modern healthcare, and community/street lighting, irrigation, waste management.

² Wood, cow dung, and crop residue for basic fuel.

³ <https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health>.

under the purview of energy poverty – (a) people who do not have access to energy though they can afford it; (b) people who have access to energy but cannot afford it; (c) all those who have neither accessibility nor affordability. The people in these groups are categorised as energy poor with group (c) as the most impoverished.

There is a need to accurately measure the extent of energy poverty (by capturing all the groups (a), (b) and, (c)) especially in the severely stricken ASEAN countries to enable effective policy measures to eliminate the problem. Past research has provided measures to assess energy poverty (Discussed in section 2.1) but none of the measures has taken care of group (b), which consists of those people who are not able to afford energy and are therefore energy poor. Hence, previous measures might provide a misleading magnitude of the prevailing problem. This paper has proposed a new method to measure energy poverty, which considers all the groups. We use this new method to calculate the extent of energy poverty prevailing in South and Southeast Asian countries.

Section two of this paper draws on the existing literature related to measurement of energy poverty. Based on the drawbacks of the existing measures, section three develops and describes a new approach to measure energy poverty taking care of the limitations of the earlier measures. It is followed by description and processing of data, used for this study. Section four describes and analyses the results obtained by applying new method of energy poverty measurement. It also gives the comparative analyses of the countries' energy situation. The paper then develops four practical village level case studies based on primary data in section five, which shed light on the extent of energy poverty and solutions available in those villages. The section also proposes affordable and sustainable solutions to eliminate energy poverty in remote/poor regions. Finally, policy implications and concluding remarks are placed in section six.

2. Literature review of alternative approaches to assess energy poverty

The idea of energy poverty was coined by Boardman (1991), who suggested a single indicator approach, that if energy expenditure is more than 10% of household income, the household is said to be energy poor. Based on this approach, Foster proposed a method to measure energy poverty by considering the lack of availability of energy (Foster et al., 2000) defined as the average final energy consumption of the households whose total incomes fall within 10% of the national poverty line. It is a demand-based approach based on primary survey, which becomes tedious at the macro level (Barnes et al., 2011, 2004). Pachauri and Spreng (Pachauri et al., 2004) suggested the Energy Access Consumption Matrix through which they measured the access of population to different modes of energy. Mirza and Szirmai (2010) developed the Energy Poverty Index (EPI), which combined the excess inconvenience associated with the energy mix used by households and a lack of sufficient energy to meet basic household needs. The International Energy Agency (IEA) proposed another method, the Energy Development Index (IEA, 2011), to assess the progress of a region in its transition to modern fuels and the degree of maturity of its energy end-use.

A composite measure was proposed – the Multidimensional Energy Poverty Index (MEPI) (Nussbaumer et al., 2012), according to which a person is identified as energy poor if the combination of his/her deprivations exceeds a predefined threshold. The index is the product of a head-count ratio (share of people identified as energy poor) and the average intensity of deprivation of the energy poor; and is based on micro data from household surveys. Practical Action (2012) proposed and published the Total Energy Access Method that measured the population's access to electricity, fuels, and mechanical power and it measures accessibility of households, enterprises, and communities. Besides these, an easy way adopted by international organisations (ADB, USAID, and IEA, 2011) is the Energy Access Method (EAM) that measures energy poverty based on electrification rates and access to modern cooking systems (Culver, 2017; IEA, 2017). In 2015, a Multi-tier Energy Access Method was

conceptualized, where energy access to households, communities and production units is measured separately (Bhatia and Angelou, 2015). Another concept, namely Energy Burden (EB) was suggested (Teller-Elsberg et al., 2016) where EB is defined as expenditure on energy as a percentage of income and if this exceeds 10% as per Boardman's definition (Boardman, 1991), it is considered to be energy poverty.

An in-depth overview of the measures used hitherto to assess energy poverty has revealed the limitations of these methods. The literature is remarkably diverse in terms of the accuracy of the calculations as well as in the degree of understanding of energy poverty assessment (Herrero, 2017). Most of these measures like MEPI, Energy Access Consumption Matrix, and Energy Poverty Index are comprehensive but demand very detailed primary data collection, which makes them very extensive and tedious. Besides, there can be a problem of 'cooked' data for various household data variables. Energy Development Index explains how a national energy system is maturing but not the extent of energy deprivation of households; hence, it is not a relevant tool to measure energy poverty. The Total Energy Access method needs data on the informal sector, thus making it very time consuming and tedious. The EAM and its various forms are relatively simple but it does not consider the affordability aspect. Thus, it is fundamental to identify and understand methodological challenges of different approaches before applying it to measure region wise energy poverty (Herrero, 2017). Herrero advocates multiple indicator approaches over single indicator approaches.

Therefore, we have attempted to develop a new multiple indicator method to assess energy poverty that takes care of accessibility as well as affordability and does not require primary data to calculate the extent of energy poverty.

3. Comprehensive energy poverty index: a new approach to assess energy poverty

3.1. Composite energy poverty index: the framework and methodology

We have developed a new method of measuring energy poverty by bringing in important parameters representing affordability of the population. The methods reviewed above focus on accessibility of energy and do not include availability and affordability of energy in a country, while our index tries to capture all the three. The relevant indicators are judiciously selected based on their relevance and quantifiability. To capture accessibility, two indicators at a country/area level are considered. These indicators are (i) electricity rate and (ii) access to modern cooking systems including LPG, electricity, and biogas cooking stoves. To measure the availability aspect, we consider the total energy supply available for use in a country, captured through total primary energy supply per capita (TPES_{pc}). For affordability, we considered total per capita energy consumed by end users captured through total final energy consumption per capita (TFEC_{pc}). We developed a composite index using these four indicators: electricity access, access to modern fuels and technology, total energy supply, and total final energy consumption.

The first step to create an index to determine the extent of energy poverty is to construct the 'weighted average of the four energy poverty indicators' (WAEPI). We used three different weight sets (W_1 , W_2 and W_3) to represent different scenarios (Table 1).

Table 1
Three weight sets used to measure energy poverty.

No.	Indicators	W_1	W_2	W_3
1	Total Population with access to electricity (%)	0.5	0.25	0.4
2	Access to clean fuels and technologies for cooking (% of population)	0.5	0.25	0.4
3	TPES per capita (TPES _{pc})	0	0.25	0.1
4	TFEC per capita (TFEC _{pc})	0	0.25	0.1

Table 2
Sources of data for the variables used.

Data Variables	Source of Data
Access to Electricity (Rural, Urban and total), clean fuels; Electric power consumption per capita; Fossil fuel energy consumption; Energy use per \$1000 GDP	http://data.worldbank.org
Total primary energy supply per capita and final energy consumption per capita;	www.iea.org/statistics www.aseanenergy.org/aeds , www.ec.europa.eu
Village Level Information for Case Studies	Primary survey through detailed questionnaires filled by a team of country experts and village heads

Table 3
The variables and their definitions.
Source: data.worldbank.org, ec.europa.eu

S.No	Variable	Definition
1	Access to electricity, urban population	The percentage of people living in urban areas who have electricity access
2	Access to electricity, rural population	The percentage of people living in rural areas who have electricity access
3	Total population with access to electricity	The percentage of people living in a country having electricity access
4	Access to clean fuels and technologies for cooking	The percentage of total population primarily using clean cooking fuels and technologies for cooking
5	Total primary energy supply per capita (TPES _{pc})	It is equal to indigenous production + imports - exports - international marine bunkers - international aviation bunkers ± stock changes
6	Total final energy consumption per capita (TFEC _{pc})	The total per capita energy consumed by the end users such as households, industry, agriculture and excludes the energy used up by the energy sector itself
7	Energy intensity	It is the energy per unit of gross domestic product (GDP). Being a measure of energy efficiency, high-energy intensities indicate a high price or cost of converting energy into GDP. We use the reciprocal of Energy Intensity here, given by GDP/Total Primary Energy Supply.
8	Electric power consumption per capita (Kwh)	The production of power plants and combined heat of power plants less transmission, distribution, and transformation losses and own use by heat and power plants, divided by respective country's mid-year populations
9	Fossil fuel energy consumption	The percentage of total energy consumption that originates from the burning of fossil fuels (coal, oil, petroleum, and natural gas products)
10	Energy use (kg of oil equivalent) per \$1000 GDP (constant 2011 PPP)	It is the kilogram of oil equivalent of energy use per constant PPP GDP, which is equal to indigenous production plus imports and stock changes, minus exports and fuels supplied to ships and aircraft engaged in international transport.

In the first weight set W_1 , no weight is given to TPES_{pc} and TFEC_{pc}, hence it is same as the original EAM (Practical Action, 2012; IEA, 2017; Culver, 2017) where only accessibility indicators (No.1,2 in Table 1) are given weights and 0 weights is given to availability (No.3) as well as affordability (No.4). This weight set is used to measure energy poverty with an established method based on secondary data. In W_2 , we include all the indicators of availability, accessibility, and affordability (No.1–4) and give equal weightage to all. This is because all these indicators play an important role in determining energy availability in the region. In the third weight set W_3 , relatively more weight is given to accessibility (No.1,2), that is 40% each (Total 80%), than to affordability and availability. This is because accessibility plays the most important role in explaining energy poverty. However, end consumption is as important (No.3), because it indicates how much is ultimately consumed. In addition, total energy supply also plays an important role as it shows per capita availability (No.4). Therefore, we include variables which represent affordability and availability, but with lesser weights, that is 10% each (Total 20%). In W_3 , Slight variation in the weight-shares in W_3 such as 70%–30% ratio does not impact the result significantly.

We construct WAEPI (Equation (1)) as follows:

$$\text{WAEPI}_{x, \text{year}} = \Sigma(w_1 * \text{Access to electricity}_n + w_2 * \text{Access to modern fuels}_n + w_3 * \text{TFEC pc}_n + w_4 * \text{TPES pc}_n) \quad (1)$$

Where

X = Country; n = Normalised indicator

$$\sum_{i=1}^4 w_i = 1$$

The value of WAEPI will indicate the extent of fulfilled energy needs of the population that indicates the degree of energy availability, accessibility, and affordability in a country. We obtained the Composite

Energy Poverty Index (CEPI) in the following way (Equation (2)).

$$\text{CEPI for a country} = 100 - \text{WAEPI} \quad (2)$$

Subtracting WAEPI from 100 gives a measure of the energy gap (which we call CEPI) a country needs to fill. CEPI provides a measure of energy poverty and measures the extent of gap in energy availability and affordability of a region.

3.2. Description of the data

3.2.1. Data sources

Various secondary data sources are used to collect data for electricity access, consumption in the economy, total supply and consumption of energy, energy intensity as well as data for rural and urban areas for two years, 2010 and 2014.

We used the [World Bank database](http://data.worldbank.org), the International Energy Agency (IEA), and the ASEAN Centre for Energy (ACE) database (Table 2). To develop case studies, we collected primary data through a detailed village-level questionnaire filled out by experts for two villages in Philippines, one in Thailand, and one in India.

3.2.2. Description of the secondary data

In the present study, we have used CEPI to assess energy poverty in five countries, four —Cambodia (Cam), Indonesia (Indo), Philippines (Phil), and Thailand (Thai) — are from ASEAN and the fifth is India (Ind). We selected these ASEAN countries because they are comparatively lagging behind in energy situation and India is selected because it is a huge economy with large areas still surviving on traditional sources of energy. We were unable to include Laos PDR (Poor ASEAN Country) in this study because of lack of availability of data. It was possibly a good data point to analyse energy poverty because the electricity access was only around 77% in 2014. Besides this, Japan is considered as the reference country.

Cambodia has a population of over 15 million. Its energy supply

Table 4a

The original data series for 2014 and 2010.

Source: Compiled from data.worldbank.org, www.iea.org, www.aseanenergy.org.

S. No.	Indicators	Cam 2014	Cam 2010	India 2014	India 2010	Indo 2014	Indo 2010
1	Access to electricity, urban population (%)	96.90	91.30	98.27	94.00	99.74	98.96
2	Access to electricity, rural population (%)	49.20	18.80	70.03	65.70	94.29	89.39
3	Total Population with access to electricity (%)	56.10	31.10	79.17	76.30	97.01	94.15
4	Access to clean fuels and technologies for cooking (% of population)	13.41	11.04	34.16	31.51	56.64	39.57
5	TPES/population (per capita)	0.42	0.37	0.64	0.56	0.88	0.87
6	Total final energy consumption per capita (TFEC per capita) (kg of oil equivalent)	360	318	430	390	635	608
7	Electric power consumption (kWh per capita)	271.43	144.32	805.60	642.11	811.9	634.18
8	Fossil fuel energy consumption (% of total)	30.68	29.47	73.46	71.40	65.56	67.15
9	Energy use (kg of oil equivalent) per \$1000 GDP (constant 2011 PPP)	133.45	146.79	118.26	127.00	88.36	103.70
10	GDP/TPES	2.33	2.13	2.56	2.38	4.17	3.57

Table 4b

The original data series for 2014 and 2010.

Source: Compiled from data.worldbank.org, www.iea.org, www.aseanenergy.org.

S. No.	Indicators	Japan 2014	Japan 2010	Phil 2014	Phil 2010	Thai 2014	Thai 2010
1	Access to electricity, urban population (%)	100.00	100.00	97.33	96.13	100.00	100.00
2	Access to electricity, rural population (%)	100.00	100.00	82.55	75.01	100.00	97.00
3	Total population with access to electricity (%)	100.00	100.00	89.13	84.57	100.00	99.70
4	Access to clean fuels and technologies for cooking (% of population)	100.00	100.00	44.85	43.23	75.85	71.54
5	TPES/population (per capita)	3.46	3.89	0.48	0.43	1.99	1.77
6	Total final energy consumption per capita (TFEC per capita) (kg of oil equivalent)	2298	2403	270	253	1412	1263
7	Electric power consumption (kWh per capita)	7819.7	8594.9	699.21	639.54	2539.6	2307.33
8	Fossil fuel energy consumption (% of total)	94.68	80.93	61.99	60.11	79.75	79.99
9	Energy use (kg of oil equivalent) per \$1000 GDP (constant 2011 PPP)	92.99	108.90	72.31	77.01	132.60	130.01
10	GDP/TPES (2010) IEA	14.29	11.11	5.26	5.00	2.86	2.86

facilities were heavily damaged by war. Cambodia has started its rehabilitation with the help of international development institutions. Access to modern energy is very low especially in the rural areas. Still, less than half of the population rely on traditional methods of energy. India is the seventh-largest country by area and the second most populous country. Of the 1.4 billion people in the world who have no access to electricity, India accounts for over 160 million. As of 2018, 86.5% of the 177 million total rural households and 95% of the 44.7 million total urban households have access to electricity. Indonesia is the world's largest island country and fourth most populous, with more than 17,000 islands. Access to electricity is gaining momentum and approximately 93% people have access to electricity (Indonesia Investments, 2017, Indonesia, 2015). The Philippines comprises of 7641 islands and is the 12th most populated country in the world. According to World Bank (World Bank, 2017), 12% of the population had no access to electricity in 2016 though this rate was higher at 16%⁴ in 2010. Still more than 10% of the population rely on other sources of energy for survival. Thailand is the world's 50th largest country by total area and the 21st most populous country in the world. Even though Thailand is 99% electrified (Provincial Electrical Authority of Thailand, 2018), there is an issue of energy security primarily because the main source of energy is domestic and imported natural gas, which means there is no energy diversification (International Energy Agency, 2016). We collected data for the following variables (Table 3):

The data series for all five countries is presented in Tables 4a and 4b.

3.2.3. Normalisation of data

For data analysis, it is essential that the data describing the selected indicators expressed in different units be on the same footing. This is achieved by data normalisation. Data normalisation is restructuring of a relational database in accordance with a series of so-called normal

forms to reduce data redundancy and improve data integrity. Comparison of variables should be standardised so that a single measurement is used to observe the efficiency or, the effectiveness of the decisions.

In this study, we normalise the data with reference to Japan. The choice of Japan is prudent; it scores highly on most energy metrics and there is no energy poverty. The Japan baseline for all variables is 100 for convenience. Indicator values that are greater than 100 upon normalisation indicate countries with better performance than the reference country for all the variables except energy use per \$1000 GDP. This is because higher energy use indicates lower energy efficiency, hence lower the normalised value of energy use, the better it is. We performed this exercise for two points of time, 2014 and 2010. The normalised values are presented in Table 5. The notation used for each series is - Country's short form followed by N (N represents normalised value) and Last 2 digits of the year.

4. Results, analysis and discussions

4.1. Composite energy poverty index – the results

The central task is to estimate WAEPI and derive the composite energy poverty index - CEPI. For this purpose, we considered the indicators listed in Table 5, for accessibility (indicators 3–4), availability (indicator 5), and affordability (indicator 6) of energy to the population. Tables 6 and 7 show three different values of WAEPI for all five countries at two points of time. Table 8 shows the values of composite energy poverty index (CEPI).

The results in Table 8 reveal an interesting story with different weights W_1 , W_2 , and W_3 .

- Without considering affordability (i.e. W_1), the numbers are smaller for all countries, with Cambodia suffering from most energy poverty in 2014 (65.24%), but they have experienced a significant improvement from 2010. Thailand is the least energy poor at 12.07%.

⁴ According to the World Bank (TheGlobalEconomy.com).

Table 5

Normalised Indicators for five Countries with Japan as the Reference Country.

Source: Authors' Calculations

S.No	Indicators	Phil N14	Phil N10	Indo N14	Indo N10	Thai N14	Thai N10	Ind N14	Ind N10	Camb N14	Camb N10
1	Access to electricity, urban population (%)	97.33	96.13	99.74	98.96	100	100	98.27	94	96.9	91.3
2	Access to electricity, rural population (%)	82.54	75.01	94.29	89.39	100	97	70.03	65.7	49.2	18.8
3	Total population with access to electricity (%)	89.13	84.57	97.01	94.15	100	99.7	79.17	76.3	56.1	31.1
4	Access to clean fuels and technologies for cooking (% of population)	44.85	43.23	56.64	39.57	75.85	71.54	34.16	31.51	13.41	11.03
5	TPES/population (per capita)	12.34	11.05	25.43	22.36	57.51	45.5	18.5	14.4	12.14	9.51
6	Total final energy consumption Per capita (TFEC per capita)	11.24	10.52	26.4	25.3	58.76	52.55	17.89	16.22	15	13.23
7	Electric power consumption (kWh per capita)	8.14	7.44	10.38	7.38	32.48	26.85	10.3	7.47	3.47	1.68
8	Fossil fuel energy consumption (% of total)	65.47	76.6	69.25	82.98	84.23	98.85	77.58	88.23	32.4	36.41
9	Energy use (kg of oil equivalent) per \$1000 GDP (constant 2011 PPP)	66.4	70.71	95.02	95.23	142.6	119.4	127.2	116	143.5	134.8
10	GDP/TPES (1/energy intensity) IEA	47.37	45	29.17	32.14	20	25.71	17.95	21.42	16.27	19.14

- With equal weight to all the parameters (i.e. W_2), the energy poverty index shoots up to a very high number. Equal weight exaggerates the energy poverty situation.
- With unequal weightage (i.e. W_3), we gave more importance to accessibility but did not ignore the role of affordability and availability, thus arriving at a balanced index.
- With either of the weights, Cambodia remains the most energy poor (With W_3 , its 69.6%) while India is second with 51.23% energy-poor population. The third energy-poor country is the Philippines, followed by Indonesia.
- Thailand is the least energy poor of all these countries with either weights.
- Comparing 2014 with 2010, all countries have improved in terms of energy availability, accessibility, and affordability though Cambodia's performance has shown the most drastic improvement and Thailand has shown less significant improvement.

All countries have improved in terms of energy availability, accessibility, and affordability though Cambodia's performance has shown the most drastic improvement, while Thailand has shown less significant improvement.

4.2. Radar charts

We used the normalised values and plotted the radar charts for all five countries (See Fig.1). These radar charts compared Cambodia, India, Indonesia, the Philippines, and Thailand to Japan. We considered the indicators in Table 5 to capture accessibility (indicators 1–4), affordability (indicators 6–8), and availability (indicators 5, 9–10).

4.2.1. Cambodia

Access to electricity and other modern energy sources: Cambodia scores poorly on access to electricity metrics. Urban areas are doing better in terms of access to electricity (Matt, 2007). However, less than 20% of rural areas have electricity. Rural electrification in Cambodia is increasing at a surprising rate.

Consumption of modern energy sources: Cambodia is a poor agricultural economy, with much less electricity access especially in rural areas. However, Cambodia has deposits of fossil fuels, natural gas and coal, fuelwood, which has played a crucial role in meeting the country's energy needs (84%)⁵ and is likely to continue to be an important energy resource for many years. Overall Consumption of energy by Cambodia is very close to that of Japan's.

Energy supply: Japan has a much larger energy supply than Cambodia.

Overall situation: Access to electricity has drastically improved since 2010. Final consumption is still low and no change has been observed over the years. Energy use per dollar GDP has also increased, indicating higher energy intensity in the country.

4.2.2. India

Access to electricity and other modern energy sources: India scores poorly on access to electricity metrics. Urban areas are doing better than most rural areas. For cooking purposes, hardly one-third of the population has access to modern energy sources.

Consumption of modern energy sources: Because of lack of access, percentage share of consumption of electricity and modern energy sources is quite less. However, overall consumption of energy including crude oil, natural gas and traditional sources like biomass is high.

Energy supply: Total energy supply per capita is very less compared to that of Japan. India's population is huge and the production is inadequate relative to the population size. Japan's GDP is much larger than India's that the positive TPES differences (in Japan's favour) are outweighed, resulting in a lower GDP/TPES for India.

Overall situation: We observe that access to electricity in urban areas is at the same position in both points of time. Rural electrification has improved. Energy use per dollar GDP has improved quite significantly, thus, implying higher energy intensity, which means improved capital intensity.

4.2.3. Indonesia

Access to electricity and other modern energy sources: Indonesia scored well on access to electricity metrics, with 97.7% of its population having access to electricity. However, less than 60% have access to safe methods of energy to prepare food.

Consumption of modern energy sources: Indonesia's consumption of energy is very close to Japan's consumption thus implying that it is mostly self-reliant.

Energy supply: The total per capita energy supply is very low compared to Japan's levels of energy sources, i.e. TPES per capita is very low. Similarly, for GDP/TPES, Japan's GDP is much larger than Indonesia, hence, making the figure for Indonesia very small.

Overall situation: We observed a significant change in the access to modern fuels for cooking. This implies a better quality of life especially in terms of health and education. Electricity access has improved in the rural areas, and energy per use and GDP have increased slightly.

4.2.4. Philippines

Access to electricity and other modern energy sources: The Philippines scored highly on electricity access, with 97% of urban inhabitants having access to electricity. Rural access to electricity, however, is poor compared to Japan. The Philippines has many remote islands, which are difficult to connect to a main grid. Filipinos have lesser

⁵ www.un.org/esa/agenda21/natlinfo/countr/cambodia/energy.

Table 7
WAEPI for Thailand, India, and Cambodia with three Different Weight Series.
Source: Authors' Calculations

Indicators	WAEPI Thai 2014 W ₁	WAEPI Thai 2014 W ₂	WAEPI Thai 2014 W ₃	WAEPI Thai 2010 W ₁	WAEPI Thai 2010 W ₂	WAEPI Thai 2010 W ₃	WAEPI Ind 2014 W ₁	WAEPI Ind 2014 W ₂	WAEPI Ind 2014 W ₃	WAEPI Ind 2010 W ₁	WAEPI Ind 2010 W ₂	WAEPI Ind 2010 W ₃	WAEPI Camb 2014 W ₁	WAEPI Camb 2014 W ₂	WAEPI Camb 2014 W ₃	WAEPI Camb 2010 W ₁	WAEPI Camb 2010 W ₂	WAEPI Camb 2010 W ₃
Total Population with access to electricity (%)	50.00	25.00	40.00	49.85	24.93	39.88	39.58	19.79	31.67	38.15	19.08	30.52	28.05	14.03	22.44	15.55	7.78	12.44
Access to clean fuels and technologies for cooking (% of population)	37.93	18.96	30.34	35.77	17.88	28.61	17.08	8.54	13.66	15.76	7.88	12.61	6.71	3.35	5.37	5.52	2.76	4.41
TPES/population (per capita)	0.00	12.79	5.12	0.00	11.38	4.55	0.00	4.11	1.65	0.00	3.60	1.44	0.00	2.70	1.08	0.00	2.38	0.95
TPEC per capita	0.00	14.69	5.88	0.00	13.14	5.26	0.00	4.47	1.79	0.00	4.06	1.62	0.00	3.75	1.50	0.00	3.31	1.32
WAEPI	87.93	71.44	81.33	85.62	67.32	78.30	56.66	36.92	48.77	53.91	34.61	46.19	34.76	23.83	30.39	21.07	16.22	19.13

Table 8

Composite Energy Poverty Index for the year 2014 & 2010 (with different weights).

Source: Authors' calculations

Country	CEPI (2014) With W ₁	CEPI (2014) With W ₂	CEPI (2014) With W ₃	CEPI (2010) With W ₁	CEPI (2010) With W ₂	CEPI (2010) With W ₃
Philippines	33.01	60.61	44.05	36.1	62.66	46.72
Indonesia	23.18	49.33	33.64	35.43	43.18	41.9
Thailand	12.07	28.56	18.67	14.38	32.68	21.7
India	43.34	63.08	51.23	46.09	65.39	53.81
Cambodia	65.24	76.17	69.61	78.93	83.78	80.87

access to modern fuels and technologies for cooking compared to Japan.

Consumption of modern energy sources: The energy composition of the Philippines consists of less fossil fuels than Japan. On the other hand, energy consumption per \$1000 GDP is two thirds of Japan's consumption level.

Energy supply: The Philippines greatly lags with respect to energy supply. $TPES_{pc}$ is quite low compared to Japan's though a lot had improved in the last 2–3 years. In electricity, the country's total installed capacity for 2016 grew to 21,423 MW compared to 18,765 MW in 2015. However, the per capita availability is still comparatively low. For GDP/TPES, Japan's GDP is much higher than the Philippines's GDP and hence it outweighs the differences in TPES.

Overall situation: The energy situation has improved over the years and access to electricity in the rural areas has increased. The energy use has improved with time though the increase is minor.

4.2.5. Thailand

Access to electricity and other modern energy sources: Thailand scored highly on access to electricity metrics. It is 100% electrified. Nevertheless, Thailand still lags behind accessibility to modern cooking fuels and technologies. A big chunk of the population (approximately 20%) still relies on traditional sources of energy for cooking needs.

Consumption of modern energy sources: Besides, low rate of modern cooking fuels, consumption of other sources of Energy is less than that of Japan's. They are self-reliant in electricity but import crude oil and gas from other countries.

Energy supply: TPES is almost half of Japan's but it is much better than other countries in this study.

Overall situation: Thailand has almost universal access to electricity in 2010 and 2014 (Shrestha et al., 2008). The total energy consumption per capita has increased considerably over time and the same can be said for energy use per dollar GDP. However, Thailand is lagging behind when it comes to access to modern energy, but this is improving over time.

4.2.6. Energy consumption in the five countries

In general, modern energy sources, especially electric power consumption, are much lesser than those of Japan. As Fig. 2 shows,

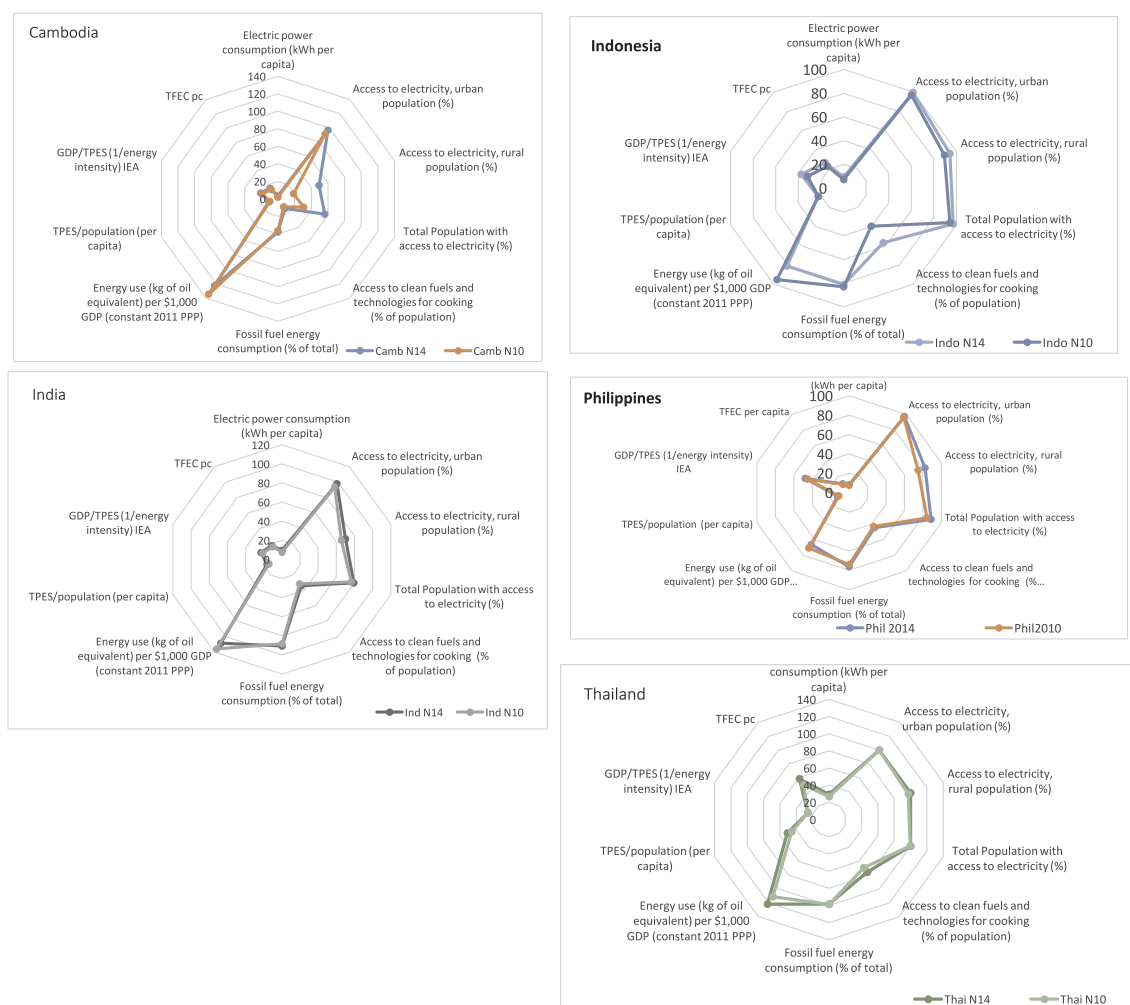


Fig. 1. Energy Indicators Radar Charts for the five Countries with Reference to Japan.

Source: Compiled from www.iea.org, data.worldbank.org, www.aseanenergy.org.

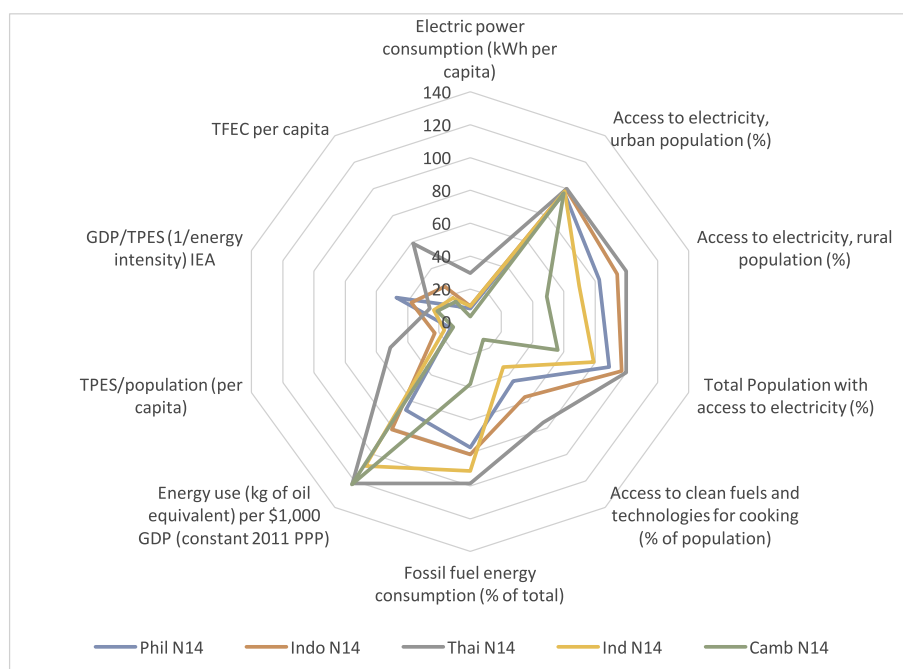


Fig. 2. A comparative chart of all Countries (2014)

Source: Compiled from www.iea.org, www.worldbank.org, www.aseanenergy.org.

Table 9

The demographic profile of all four villages.

Category	Cha, India	Cabra Island, Philippines	Carabao, Philippines	Pha Daeng, Thailand
Total population	279	2159	10, 825	219
Total males	169	1093	4725	126
Total females	110	1066	6100	93
Adult males	119	892	3365	101
Adult females	70	908	4325	78
Male children	50	201	1360	25
Female children	40	158	1775	15
Number of households	28	481	2176	41
Average size of household (Members)	6–7	5	5	5.34
Household composition (Proportion of adults to children)	2.1	2.3	2.45	5.25
Annual income level per household	US\$1000	US\$ 3458* (PhP180,000)	US\$4657 (PhP242,400)	US\$640 (20,000 baht)

Source: Questionnaires filled by respective country experts.

* US\$1.00 = PhP52.056 (Bangko Sentral ng Pilipinas) used in both the questionnaires filled by Philippines officials.

regarding electricity, Thailand is doing better than most of the five countries in most aspects, while Cambodia is at the bottom.

Cambodia is the worst in terms of access to electricity, access to modern energy sources, availability and consumption of modern energy. However, Cambodia performed well on electrification access as the radar chart revealed huge improvements between 2010 and 2014, especially the rural population (from 18% to 49.2%). Access to modern energy sources in most of these countries remain poor and there were no visible improvements between 2010 and 2014 in most countries. Indonesia is one notable exception that has seen improvements in access to clean technologies for cooking.

For all countries, fossil fuel energy consumption as a percentage of total energy consumption has remained constant or has slightly increased. This is consistent with our view that these countries are more concerned about meeting basic electricity needs than carbon emission reduction or climate change issues. However, India is the biggest of all these countries; hence, the absolute number of people getting affected by energy poverty is much more. Overall, for most countries it is still a long way to go before energy poverty is eradicated.

5. The four case studies to analyse the ground level situation

CEPI and the comparative cross-country analysis in section 3 describes the energy situation at a macro level for the five countries. Nevertheless, it is important to shed light on the ground level energy situation, which would enable us to have insights on the reasons behind region-specific energy poverty that is missing in a macro level analysis. This is particularly important because it would enable the policy makers to recommend effective solutions to eradicate energy poverty. Therefore, we developed four village level case studies taken from three out of the five countries, which aimed to supplement the macro-level empirical analysis with micro-level data.

We covered four villages, one from India, two from Philippines, and one from Thailand, to analyse the micro-level situation. These villages are Cha Village, India, Cabra Island & Carabao Island, Philippines, and Pha Daeng, Thailand. The basic demographics of these villages are provided in Table 9.

5.1. Cha Village, Ladakh, India

5.1.1. Background

Geographical and Climatic Details: Cha Village is located in a very remote area in Ladakh, Jammu and Kashmir, amidst the Himalayas in Northern India. It is located 14,000 feet above sea level, and has extremely cold weather. Hence, the villagers need lot of energy to heat up their homes.

Energy Situation in Cha: Until few years back, the village was dependent on kerosene,⁶ wood, and yak dung to fulfil its energy needs. Due to its tough geographical location, it was not possible to connect this small village to the main electrical grid. Hence, people were solely dependent on kerosene for their electricity requirements. For cooking, they use *chulha* in which yak dung is used as the prime fuel. However, there are severe health hazards associated with its burning though houses are equipped with traditional chimneys. Wood is used to heat up homes in extreme cold temperatures, which is either collected from jungles or bought from a local wood shop. Evidently, fuels used were insufficient, costly, and led to health vulnerabilities. Cha villagers had to spend about 15% of their income for fuels for lighting and other energy needs. A combination of poor geography and a lack of money drove energy poverty in this village.

5.1.2. A successful attempt to eliminate energy poverty

A few years back, the villagers were introduced to individual solar panels and almost every villager obtained one solar home lighting system. The earlier solar home lighting provided only 1–30 W compact fluorescent light (CFL).

In 2015, Global Himalayan Expedition (GHE)⁷ built and installed 10 260 W (based on 24 V) solar DC-based microgrids in the village. These microgrids operate independently and can tie up with the grid in the future, if grid reaches the village. GHE supports the microgrid with its own funding, maintenance, and management system. GHE relies on funding from village tourism, corporate social responsibility (CSR) schemes and selling devices to the villagers. GHE improves lives by both increasing income and providing electricity.

The levelized cost of electricity (LCOE) from this system is 10 cents per kWh. GHE charges between US\$1.5 to \$2.5 per every month depending on the size of the microgrid. Additionally, the community is charged US\$2 per year for maintenance of the microgrid. Overall, the households are spending US\$20–US\$30 yearly as expenditure on electricity. This is just 3% of the yearly household income, which is very affordable. Overall, the percentage contribution of solar microgrids in fuel consumption is 90% and traditional fuels is 10%.

5.1.3. Benefits

- Due to electricity, each house has lighting, a mobile charging/a plug-in point. Out of 28 households, 10 have a television set.
- Street lighting is provided to the village with two solar street lights (SSLs) supported by each microgrid with a total load of 24 W for street lighting.
- A public health facility is also built and facilitated with a solar microgrid, which consumes 70 W of electricity for its needs.
- Farming activities have improved due to outdoor lighting through SSL.
- The average income level increased, and the youth has started returning to the village as facilities have considerably improved. Electrification has enhanced tourism and has provided livelihood to the native villagers in a big way.
- Education has also improved due to extra hours of study available now.

5.2. Cabra Island, Philippines

5.2.1. Background

Geographic and Climatic Details: Cabra Island is located in Lubang, Province of Occidental Mindoro, Philippines which is characterised by two pronounced seasons: dry from November to April and wet during May to October.

Energy Situation in Cabra Island: Cabra Island is not connected to a power grid. Given the archipelagic nature of the Philippines, the island is categorised as an off-grid. Considering the limited economic activities in the area, interconnection of Cabra Island to the main grid is presently deemed uneconomical. For cooking needs, it relies on solid fuels, i.e. biomass and firewood. Either people spend PhP100 per day on wood or the family members collect it from nearby areas, spending 3–4 person-hours per day. Moreover, 40%–50% of the households have gas stoves, which run on kerosene that creates indoor air pollution and none of the houses have a chimney to get rid of the pollution. Women and young children have the highest exposure to pollution. Kerosene is also used for power generation and transport purposes.

5.2.2. An attempt to eliminate energy poverty

Recently, the National Power Corporation (NPC), a government-owned organisation mandated to provide electricity to remote areas, developed and installed a 3×75 KVA diesel-powered microgrid to provide electricity on the island. Out of 481 households, 461 get 8 h of electricity from this microgrid, which means 95.8% of households have access to electricity. They spend 10% of their income on electricity bills, if subsidies are considered. Microgrids ensure that they receive light, basic cooling, mobile charging point, and other comforts in their houses, thus, improving the quality of life (Table 10). Besides households, the microgrid has enabled 25 street lights in the village, which are mostly unmetered (Table 11). Electrification in Cabra Island has led to increased comfort and leisure levels. Five percent of the households even have computers.

5.2.3. Benefits

- Due to electricity, 95% of the houses have lighting, and various plug-in points for electrical devices.
- Street lighting is provided to the village with 25 units covering all the streets. They are also supported by the microgrid. The load of a 1-m light is 121 kWh, while the other 24 lights have a load of 15 kWh, resulting in a total load of 360 kWh for street lighting.
- Microgrid is also providing electricity to one barangay health centre, one elementary school and a high school.
- Due to electricity, people live in safer, healthier, and more educated environments. They can focus on their work, schools, and health, thereby improving productivity and income levels.

5.3. Carabao Island, Philippines

5.3.1. Background

Geographic and Climatic Details: Carabao Island is located in San Jose, Romblon in the Philippines and has a tropical climate. The island has a diverse geographic profile, with irregular coastline interiors characterised by undulating hills covered with rainforest and coconut trees.

Energy Situation in Carabao Island: Carabao Island is an off-grid area that is yet to connect to the national transmission system. At present, the limited economic activities in the area has discouraged a connection to the main grid. Hence, they have four diesel microgrids installed by the National Power Corporation, which caters about 85% of the households. Besides this, some houses (exact number not known) have independent solar panels on their rooftop.

For cooking needs, these villages use firewood. They spend PhP 25 per day or PhP 9125 per year to buy wood, which is 3% of their annual

⁶ A fuel obtained from petroleum.

⁷ A registered Non-Governmental Organisation, which is electrifying villages in remote areas of Himalayas.

Table 10

Percentage of houses using different electrical devices.
Source: Questionnaire filled by Country Experts

Appliance	No. of Houses Using the Appliance (Estimated, %)
Television	50
Refrigerator	20
Sound system	20
Welding machine	1
Computer	5
Electric fan	50

Table 11

Street lights in Cabra village.

Source: Questionnaire filled by Country Experts

Type	No. of Units	Load
Metered Street Lights	1	121 kWh
Unmetered Street Lights	24	15 kWh

income.

Overall, the families spend 13% of their income on obtaining traditional and modern energy. According to Boardman (1991), families spending more than 10% are considered energy poor.

5.3.2. A successful attempt to eliminate energy poverty

The NPC provided four microgrids to power the 2176 Carabao Island households successfully electrifying 85% of households on the island. According to our CEPI index, only 15% of the population suffers from energy poverty, though according to Boardman's definition, the island is still energy poor.

These microgrids have a total load of 904 kW, providing the island with 16 h of electricity service per day (Table 12).

The islanders still use only 10%⁸ of their income to pay electrical bills (after considering subsidies) as Carabao islanders earn higher incomes.

These microgrids have enabled the use of many electrical devices in these households. Table 13 shows that 98% of houses have access to cell phones, a crucial device for communications, and 60% have television sets. Compared to Cabra Island, these numbers of Carabao Island are much larger, showing more penetration of these devices.

The larger penetration in Carabao Island could be due to its size. A bigger market for electricity has led to better microgrids to provide electricity to the homes on the island. Population size might be an important factor when it comes to electrification.

5.3.3. Benefits

- A large proportion of households (85%) have been successfully electrified not only with basic lighting but are also enabled to use various electric appliances and gadgets that improve standard of living.
- Two schools and one hospital had been electrified thus providing better quality education, as well as enabling better health facilities respectively.
- Electricity leads to modern and more efficient methods of farming, which will improve the yield and, and more non-farm jobs leading to income levels.
- Access to electricity also creates more jobs for the youth because of increase in commercial activities on the island.

Table 12

Load of the four diesel microgrids in Carabao island.

Source: Questionnaire filled by Country Experts

Unit	Load (kW)
1	163
2	163
3	228
4	350

Table 13

Percentage of houses using electrical devices.

Source: Questionnaire filled by Country Experts

Electrical Device	Percentage of Houses using it (Estimated, %)
Television	60
Refrigerator	50
Computer	10
Electric fan	80
Cell phone	98
Air-conditioner	10
Rice cooker	10
Washing machine	10

5.4. Village Pha Daeng, Phayao, Thailand

5.4.1. Background

Geographical Details: Pha Daeng Village is located in Phayao Province in Doi Luang National Park, Northern Thailand. It lies in a cold mountainous region climate and hence, and requires constant heating of the house.

Energy Situation in Pha: The primary sources of energy in Pha Daeng village are (i) solid fuel (firewood); (ii) gas fuel (LPG); (iii) liquid fuel (kerosene); and (iv) electricity through independent solar panels over rooftops. For cooking, wood and LPG are the primary sources. Only 10% of the houses have gas stoves with LPG (modern energy source) and the rest depend on wood, which is a traditional fuel. The village is not connected to the main electricity grid because it is located in Doi Luang National Park area, which is regulated under national law which that prohibits main grid installation in the area.

Pha Daeng village is situated in a cold, region and, therefore need heating devices for thermal comfort. The villagers kindle firewood during winter and family members spend on average 4 h a day collecting wood from nearby jungles.

5.4.2. An attempt to reduce energy poverty

In 2006, the village was partially electrified by solar panels provided by the government, free of charge. Twenty-four households have solar panels, which enable villagers to use lighting, television and charging mobile phones. Each of these 24 solar panels has a capacity of 125 W, and consists of a solar module, solar charge controller, battery, and inverter. Besides households, the local medical clinic is electrified through a solar panel, which is maintained by the clinic staff. A no-cost model also means that children under the age of 14 years do not have to work to collect fuel.

However, the remaining houses rely on kerosene lamps for lighting. The village does not have street lights and schools do not have access to electricity. Many households still rely on wood for cooking and thermal needs. Overall, the percentage contribution of solar panels in fuel consumption is 20% and that of traditional fuels is 80%.

5.4.3. Benefits

- About 59% of the households have electricity to fulfil their needs.

⁸ The threshold recommended by international standards.

- They are able to have lights, a television set, and a charging point.
2. A solar panel also electrifies a public health facility.
 3. Education has also improved due to extra hours of study available for children of the households who have LPG and solar panels for their energy needs.

5.5. Lessons learnt from the case studies

In Cha village, the installation of these solar microgrids have eliminated energy poverty from this village and has enhanced livelihood through tourism and other productive activities, along with fulfilling electricity needed for schools, health centres, and streetlights. This is a successful example of a greener way to provide electricity access to remote areas.

In Cabra and Carabao Islands, the energy situation has greatly improved with diesel-operated microgrids installed by the government thus alleviating a large proportion of energy poverty. Microgrids are a growing segment of the energy industry, representing a paradigm shift from central station power plants towards more localised, distributed generation, especially in remote areas. Currently, the Cabra and Carabao villages relies on a fossil fuel-based microgrid. Until now, no energy resource assessment has been conducted in Cabra Island to identify potential renewable energy resources in the area. Electrification in these Philippines villages has helped reduce the time allocated to firewood collection by household members, improves the labour supply of men and women, schooling of children, household per capita income and expenditure, and overall quality of life.

Solar power in Pha Daeng village, Thailand, has proven as a viable alternative to an electrical main grid for more than 50% of households. However, the solar infrastructure needs to be expanded to provide street lights, electricity to the ‘un-electrified households’ and to enable households to have heating devices so that they do not have to rely on firewood for thermal comfort.

Because the independent solar panels in few houses cannot solve the problem of access to modern energy for the entire population, the Thailand government has recognised the need for electricity and, hence, has authorised the Ministry of Energy to evaluate the feasibility of installing micro-grids in the village. Microgrids seem to be a great alternative that circumvent legal requirements and geographical challenges and provide sufficient electricity. They could cater to larger electricity needs and improve the overall quality of life. They would also lead to more productive activities in the village, which in turn would provide better economic prospects and higher income levels.

5.6. Proposed solutions to eliminate energy poverty based on case studies

Based on the above case studies, we found that a major share of population in rural and remote areas still lack electricity and modern cooking fuels. Hence, to solve the electricity issue and keeping in mind that green solutions are the future, we propose solutions to provide electricity in remote areas where electricity is not connected. Different-sized microgrids are a very efficient solution for rural, remote, and poor areas. Table 14 suggests different sizes of microgrids and their attributes, while Table 15 suggests microgrids according to population size.

Besides providing solutions to deliver electricity, we have to make necessary inroads on clean cooking because the problem impacts so many segments— energy poverty, health (Zhang et al., 2019), gender, and environment. The governments face a big challenge to intervene, set up policies and take urgent steps to provide efficient cooking stoves, which run on LPG or natural gas, at affordable prices. It is important and urgent to reduce the population's dependence on traditional fuels like firewood, charcoal and kerosene for cooking (Mulqueen and du Pont, 2014). Moreover, people are used to traditional sources like firewood or animal dung, which are available at a negligible financial cost. Therefore, it is a challenge to persuade this section of the population to spend on cleaner fuels. It is a big task for the authorities to

Table 14
Proposed size-based solutions to electrify villages.
Source: Authors' recommendation

Type	Stand-alone systems	Pico Grid	Nano Grid	Micro Grid
Size	0–0.1 kW	0–1 kW	0–5 kW	5–100 kW
Features	Emergency power systems	Single load systems	Single voltage	Manage local energy supply and demand
Technology	Solar based	Solar, battery, diesel	Hybrid solar, wind, diesel, battery	Provide variety of quality and reliability options
Mode	Off grid	Off grid	Off grid, grid connected	output energy systems
Load profiles	1 CFL/LED lights	Light, mobile charging point, SSL, Low Voltage, Direct Current Appliances	DC power devices television, computer, refrigerator, fans, pumps & motors, street lights	Varying pricing possible Multi-hybrid solar, wind, diesel, hydro, biomass, battery Off grid, grid connected Air conditioning, heaters, industrial, community hospitals, educational institutions, schools

Table 15

Proposed solutions to electrify villages according to population size.

Source: Authors' Recommendation

Socio-economics	Stand-alone Systems	Pico Grid	Nano Grid	Micro Grid
Average population	3–5 persons	250–300 persons	1750–2500 persons	Up to 10,000 persons
No. of households Supported	1–2 max	25–30	450–500	Up to 2000
Electricity bill % of income		2%–3%	3%–4%	4%–5%
Annual power consumption	30–50 KWh	300–400 kWh	500–750 kWh	1200–1500 kWh
Levelized cost of electricity		8–10 cents/kWh	11–12 cents/kWh	13–15 cents/kWh
Annual income levels	US\$300–400	US\$600–1000	US\$1500–2500	US\$4000–5000
Business model	Donors	CSR, NGO, Sponsors	Government Subsidised	Government Subsidised
Clean energy mix	100%	80:20 Mix	60:40 Mix	70:30 Mix
			Diesel still used as dominant off-grid solution	Diesel used as standby solution for intermittencies

work out a low cost solution for clean cooking.

6. Conclusions and policy implications

This paper has proposed a new method, CEPI, for assessing energy poverty by including indicators that cover the aspects of availability and affordability, which suggest a lack of energy sources in daily life. CEPI is calculated with different weights for different indicators. We experimented with altered weights, ranging from 0 to 0.4. The results revealed that though the intensity of energy poverty becomes severe with change in weights, the ranking of countries in terms of depravity remains the same. This suggests that the method is robust.

Results suggests that Thailand remains the least energy poverty-stricken while Cambodia, is the most energy poverty-stricken. Besides Thailand suffering least from energy poverty, it has a relatively good energy position and has the highest rate of improvement in energy reach-out to the population. Cambodia remains the most energy poor (70%) while India is second, with 50% energy-poor population. Followed by the Philippines and Indonesia. Comparing 2014, 2010 with radar charts, all countries had improved in terms of energy availability, accessibility, and affordability. Cambodia's performance has shown the most drastic improvement, while Thailand has exhibited less significant improvement.

Further, the four case studies enlightened us on the energy situation in rural areas of three countries where attempts have been made to eliminate energy poverty by bringing in low-cost solutions like micro-grids or solar panels. Except for, in India, government has played a primary role in bringing in these solutions. The presence of a sustainable energy poverty solution provides the population with a comfortable life, employment opportunities, and a better livelihood (Pereira et al., 2018). Electricity delivers a foundation for progress of human communities and reduces the requirement for traditional sources of energy, such as wood, animal waste, and crop residues (Goldemberg, 2000). Women and children achieve better living conditions due to reduced indoor pollution, less time spent in collecting fuel, and improved school attendance. These case studies have revealed the possibility of adopting practical solutions, which can reach remote and poor areas.

Effective policy measures are required to kick-start the eradication of severe energy poverty. Policies already exist on papers in most countries but a rigorous and effective implementation of these policies is required. Besides policies, a supportive regulatory and leadership role of the government is also important, as huge investment is required to supply electricity and safe heating energy sources. For instance, a subsidy-based tariff model is required for very poor communities with a subsequent reduction in subsidies, when the production of electricity becomes efficient. For investment, public-private partnerships are a good solution with both sharing their technical expertise, market and resource evaluation, access to financing, and project implementation.

Besides policies, it is important to develop cheap and sustainable technologies for these societies. This paper suggested solutions to cater

to the needs of rural areas depending on the needs and size of the village. These solutions range from a stand-alone system of 0.1 kW capacity, catering to a single household, to a large microgrid (up to 100 kW) catering to an entire village. These portable energy providers run on solar, wind, or a hybrid with a diesel and a battery back-up. They have a low LCOE; hence, when passed on to the consumer in form of tariffs, the electricity bill is quite low in proportion to the total income of the household. They can run on a single voltage or a variety of voltage depending on the size and need of the community. With this kind of energy provided, people can possibly have basic lighting, plug-ins, and equipment required for small commercial enterprises. A keen government can easily adopt these low-cost and sustainable solutions for secluded and less dense off-grid communities.

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